TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

Proposer	Name	Qing Cao	Email	qingcao2@illinois.edu
Institution / Department	University of Illinois Urbana-Champaign, Department of Materials Science and Engineering			
Address	104 S. Goodwin Ave., Urbana, IL 61801, USA			

2. Technology Overview

Title of Proposed Technology	A surface technology killing bacteria in contact without releasing chemicals			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Bio - Biomaterials	Fine Chemicals	Nano-Application Technologies
Concept of Proposed Technology	The Covid-19 pandemic and consumers' increasing concerns over hygiene are driving the demand for antimicrobial coatings and packaging, with the global revenue projected to $23.08 billion by 2033. However, the current commercial antimicrobial films, which function mainly based on the release of toxic biocides and antibiotics, suffer from very limited antimicrobial activities in real use environments and could incur risks of environmental contaminations, drug resistance, host-tissue toxicity, and acute inflammatory responses. We have recently identified a unique polymer antimicrobial nanostructure with optimized geometries, which mimics the nanoprotrusions found on the surface of cicada wings. Such biomimetic mechano-bactericidal nanopillar arrays are capable of effectively killing a wide spectrum of attached bacteria through a purely physical process without directly releasing any chemicals or harming mammalian cells. They can eliminate >99% of bacteria and prevent the formation of biofilms both *in vitro* and *in vivo*. This project will then focus on **(1)** establishing the scalability to manufacture such bioinspired physical antimicrobial foils, **(2)** drastically reducing the material cost and improving sustainability by constructing the films based on low-cost, biodegradable polymers, and **(3)** performing systematic benchmarks of antimicrobial performances against commercial chemical-based antimicrobial films to establish factual basis for marketing and regulatory clearance.			
Rationale for Joint R&D Project	A strong rationale for a joint R&D project between the university and a SME centers on the complementary strengths each partner brings to the development and commercialization of next-generation physical antimicrobial materials. The university contributes deep scientific expertise in biomimetic nanostructures, polymer engineering, and mechanobiology, along with access to advanced fabrication and characterization facilities needed to refine the cicada-inspired mecahno-bactericidal nanopillar technology. Meanwhile, the Korean SME offers the market-oriented			

	engineering capabilities, manufacturing know-how, and market awareness required to translate laboratory-scale discoveries into scalable, cost-effective products suitable for packaging, medical, and consumer-hygiene markets. By combining the university's innovation in physical antimicrobial mechanisms with the SME's strengths in process scaling, equipment development, regulatory navigation, and product deployment, the partnership accelerates the path from scientific discovery to commercial impact.
Keywords	1. Antimicrobial 2. Bactericidal surfaces 3. Biomimetic 4. Packaging 5. Roll-to-roll polymer manufacturing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3/4)	**End of the Project**	Level (6/7)

R&D Objectives

The Societal Need and the Customer: Antimicrobial films can inhibit the growth and spread of microorganisms on surfaces. The global market for antimicrobial packaging was valued at $11.21 billion in 2023 and is projected to grow to $23.08 billion by 2033 at a compound annual growth rate (CAGR) of 7.49%. They are primarily used in the healthcare and food industries to prevent hospital acquired infections and extend the shelf life of packaged foods. The commercial antimicrobial films function through the slow release of toxic biocides such as heavy metal ions, antibiotics, and nanoparticles. They have limited antimicrobial performance especially over extended period of time, raise toxicity concerns, and induce drug resistance.

Biomimetic mechano-bactericidal nanostructures: the foundation of the proposed project.

Our previous research offered an approach to potentially overcome these limitations. Several naturally occurring bactericidal surfaces have recently been discovered. For example, cicada wings can kill *Pseudomonas aeruginosa* within 3 min of contact, based on purely physical interactions. Cell corpses are subsequently released as debris in 20 min. Examining surfaces of cicada/dragonfly wings revealed regular arrays of nanopillars (**Fig. 1A**). When bacteria contact these nanopillars, regions of cell membrane are mechanically ruptured to causes cell necrosis (**Fig. 1B**). These mechano-bactericidal nanostructures, which have topologies distinct from superhydrophobic surfaces (~50 times smaller diameter and pitch, and 3–4 times higher aspect ratio), might promise an unprecedented broad-spectrum antimicrobial effectiveness without toxic side effects. We have three accomplishments which laid the foundation for the proposed translational activities with our SME partners.

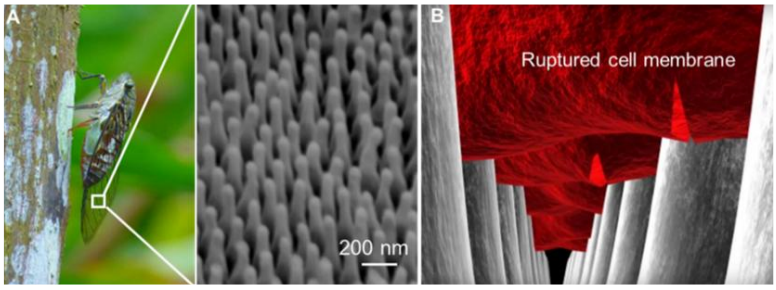

Fig. 1. Bactericidal cicada wings. (A) Optical image of a cicada with a micrograph of its wing showing the nanopillar arrays. (B) The mechano-bactericidal mechanism.

(1) Justification of the optimum biomimetic designs on polymer foils.

The controlled formation of high-density and high-aspect-ratio nanostructures mimicking these actively mechano-bactericidal surfaces found in nature was a significant challenge, which prevented

us from justifying the best biomimetic design for envisioned antimicrobial applications. In our previous works, we developed a process combining nanosphere lithography and anisotropic etching to fabricate nanowell templates with precisely and separately defined pitch, diameter, and depth, on single-crystalline silicon wafers. They were then used to prepare ~1×1 cm^2 polyimide foils featuring high-density surface nanopillars. Specifically, we started with a single-crystalline silicon wafer covered by a layer of oxide as substrate (**Fig. 2A**). Monodispersed polystyrene nanospheres were deposited by spin casting. Upon drying, nanospheres self-assembled into a hexagonal-close-packed monolayer (**Fig. 2B**). A timed oxygen-plasma reactive ion etching (RIE) then trimmed down sphere size and created controllable inter-sphere gaps (**Fig. 2C**), which was covered by deposited 10 nm-thick Cr thin film. After metal deposition, the colloidal crystal mask was removed in chloroform, followed by etching of the exposed SiO_2 by CF_4 RIE, with the Cr film as the etching mask (**Fig. 2D**). Deep-Si RIE finally created high-aspect-ratio nanowells, using the patterned Cr/SiO_2 as the etching mask (**Fig. 2E**). In this process, *the nanowell pitch, diameter, and depth were all precisely and separately controlled by the starting polystyrene nanosphere size, etching time of O_2 RIE, and the number of etching cycles in deep-Si RIE, respectively*. After obtaining master templates, polyamic acid dissolved in N-methyl-2-pyrrolidinone was coated onto the template by doctor blading. Vacuum annealing converted oligomers into polyimide that can be peeled off from the template as a standalone film, featuring high-density nanopillar arrays on its surface (**Fig. 4F**).

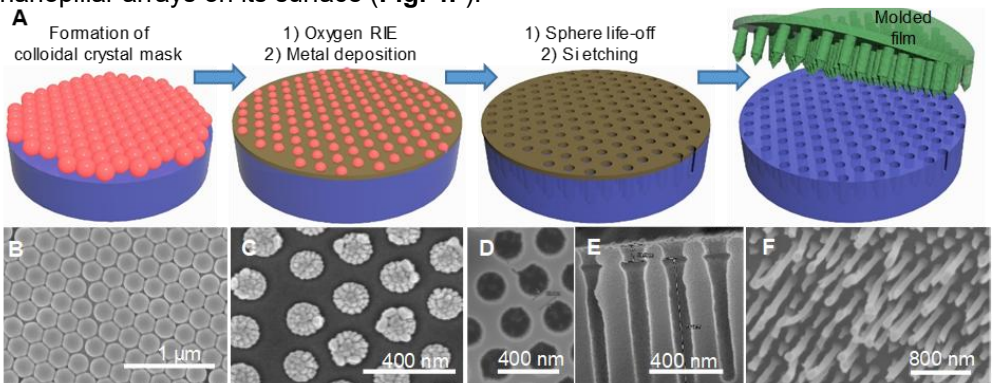

Fig. 2. Our current process to fabricate nanopillar arrays on Kapton films. (A) Schematics showing the process flow. Top-view SEMs of the colloidal crystal mask before (B) and after (C) O_2 reactive-ion etching (RIE), and the metal mask formed after lift-off (D). (E) The cross-sectional SEM of the template after deep Si RIE. (F) The tilted-view SEM showing the optimum nanopillar structure eventually formed.

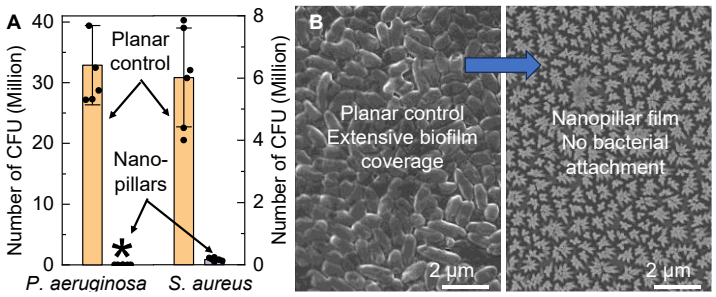

Fig. 3. Antimicrobial performance. (A) Eliminating *P. aeruginosa* and *S. aureus in vitro*. (B) Preventing *P. aeruginosa* biofilm formation after incubation.

We have prepared films with 16 different topologies. *A systematic mapping between surface nano-topologies and their quantitative antimicrobial activities has been established and the optimum nanopillar structure was determined to be around 200 nm pitch, 100 nm diameter, and 400 nm height for polymers with modulus in the range of 1–10 GPa*. Briefly, the nanopillar pitch needs to be smaller than the size of bacteria to maximize contact. The nanopillar diameter needs to be small enough to cause sufficiently large mechanical stress upon contact with the bacterial envelope, while the nanopillar aspect ratio needs to be controlled within an optimum range to induce large cell distortion for high bactericidal efficacy but also maintain nanopillars' own mechanical robustness. With an area of 1×1 cm^2 and incubation time of 3 hours to ensure most bacteria touch the surface in their Brownian motion, *the polymer nanopillar arrays with optimum geometry eliminated virtually all viable Gram-negative P. aeruginosa* (ATCC 27853) *as well as 98-99% of viable Gram-positive S. aureus* (strain

29213) in their 40 μL liquid suspensions inoculated on top, compared to planar foils as internal controls (**Fig. 3A**). *Moreover, the foil can effectively prevent biofilm formation* after 48-hour incubation in a nutrient-rich medium (**Fig. 3B**) but still maintains good structural stability and bactericidal activity after >5 washing and drying cycles.

(2) Establishment of safety with human cells.

For commercial applications, safety with human cells is as important as the bactericidal efficacy against bacteria. In our previous works, we found that *these polymer nanopillar arrays did not affect the viability of human and mouse muscle and skin cells* after 48-hour incubations as measured by both the water-soluble tetrazolium salt (WST-1) assay and the Trypan blue exclusion (TBE) test (**Fig. 4**). Several factors could contribute to this difference. **(1)** Mammalian cells and bacteria have quite different envelopes and lipid compositions. As a result, their membranes have different capabilities to accommodate the imposed surface stress. **(2)** Bacteria maintain 10–100 times higher trans-envelope osmotic pressure, creating a critical challenge for them to undergo cell envelope remodeling on nanopillar surface. **(3)** Mammalian cells contain cytoskeleton systems, which can transfer external stress to internal structures and organelles to better preserve their mechanical integrity.

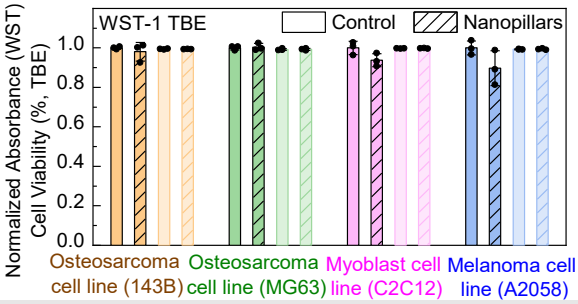

Fig. 4. Mammalian-cell viability on polymer nanopillar arrays (hatched) and planar controls (solid) determined in WST-1 (dark) and TBE (light) assays.

(3) Establishment of antimicrobial efficacy and safety in various *in vivo* tests.

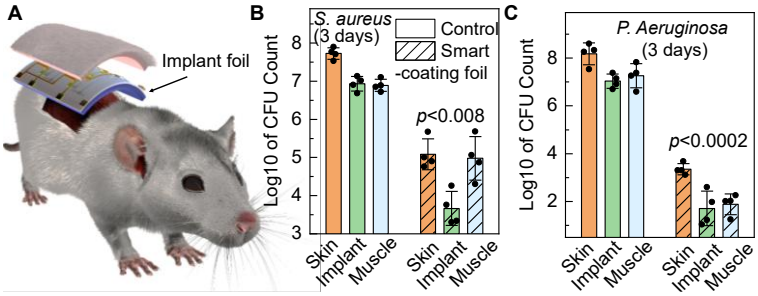

Fig. 5. Efficacy evaluated in an implant-infection mouse model. (A) Schematic. (B-C) Burdens of *S. aureus* (B) and *P. aeruginosa* (C) on implanted film and its surrounding tissues.

To fully justify the translational value and potential of this technology, in addition to these *in vitro* results, *antimicrobial performance and biocompatibility of biomimetic mechano-bactericidal polymer nanopillars have also been verified in a rodent implant infection model* (**Fig. 5A**). Here CD-1 mice (7–8-week-old, 4 per cohort) were anesthetized, and a 1 cm long incision was made through the depilated skin of the dorsum. The skin was bluntly dissected from the underlying tissue layers using iris scissors, forming a subcutaneous pocket into which the mechano-bactericidal films, or planar controls, which had been sterilized and incubated for 4 hours at 37°C with **(1)** 3×10^7 colonies forming units (CFUs) of *S. aureus* 29213, or **(2)** 3×10^7 CFUs of *P. aeruginosa* 27853 in their liquid suspensions, were placed. The number of viable microbes used far exceeded the minimal infective dose. The incision was closed with tissue glue. After 3 days, a histological examination revealed that planar films inoculated with *S. aureus* and *P. aeruginosa* induced fulminant inflammatory cell infiltrates, and bacterial micro-colonies were prevalent in deep dermis and subcutis. In contrast, polymer foils with surface mechano-bactericidal nanopillar arrays eliminated intralesional bacterial colonies and substantially reduced the extent of neutrophilic inflammation and tissue damage. Quantitatively, the numbers of CFUs of *S. aureus* and *P. aeruginosa* recoverable from the mechano-bactericidal foils and their surrounding

	tissues were orders of magnitude lower both 3 days (short term) and 2 weeks (long term) after implantation (**Fig. 5B-C**). Meanwhile, neutrophils, dendritic cells, and macrophages accumulated at the implant site quickly returned to baselines after four weeks, with the absence of T cells, as measured by flow cytometry. These results confirmed no acute immune response. **Objectives:** In our previous laboratory studies, we have justified the optimum biomimetic design of surface mechano-bactericidal nanostructures and verified their efficacy and safety both *in vitro* and *in vivo*. We have also identified the most critical obstacles to overcome together with our SME collaborators for translation. **(1) Process scalability**: The combination of silicon wafer, nanosphere lithography, and dry etching enabled us to produce templates with precisely and separately controlled nanopillar pitch, diameter, and height, which is essential for exploring and justifying the optimum biomimetic nanopillar designs. However, it incurs high material, processing, and equipment costs, and the overall size of the template is limited to the diameter of single-crystalline silicon wafers. To scale up manufacturing toward commercialization, the ***1st critical goal*** is to develop an alternative cost-effective process that can produce templates with the same optimum geometry (~200 nm pitch, ~100 nm diameter, and ~ 400 nm height) but uniformly over a much larger area on low-cost substrates. **(2) Materials cost and sustainability:** In our laboratory studies, we prepared biomimetic mechano-bactericidal nanopillars on a polyimide substrate. Polyimide, as a specialty chemical, offers advantages of flexible construction processes and good biocompatibility, and therefore it is widely used in biomedical applications. However, the cost of polyimide is high due to the high material cost of its monomers and the curing process requiring high temperature (250 ^{o}C) and nitrogen or vacuum environment. Moreover, polyimide has strong absorption in visible region, making it semi-opaque with a yellow or brownish color, which is undesirable for packaging applications. Most importantly, since polyimide has excellent thermal stability and resistance to chemicals, it is not biodegradable and will create substantial ecological problems if used in disposable packaging. For a cost-effective and environmentally friendly solution, the ***2nd goal*** is to form the optimum biomimetic mechano-bactericidal nanostructures on low-cost biopolymer foils, which are ideally derived from natural products instead of petroleum and can break down easily in soil, water, and industrial composting environments, using large-size templates and a high-throughput, continuous manufacturing process. **(3) Regulatory clearance and customer acceptance:** Beyond the proof-of-concept experiments we have already carried out, marketing and regulatory clearance of a novel type of antimicrobial foils require extensive performance evidence based on industry-standard efficacy test protocols to authenticate their advantages over existing products. Their lifetime and reliability also need to be established. ***The 3rd goal of this project*** is therefore to determine the quantitative antimicrobial efficacy of our mechano-bactericidal films against common bacteria, following the ISO protocol 22196, in comparison to commercially available alternatives. To make a stronger case for adoption by our target customers, including food manufacturers, grocery chains, hospitals, and consumers, we will also evaluate the capability of our bio-inspired mechano-bactericidal foils to reduce the surface bacterial accumulations in hospitals and food models. ***Our end goal*** is a comprehensive translational solution, including the manufacturing system optimization and ISO standards implementations.
R&D Target Specifications	Major milestones and our timeline: • 3 months: Establish protocol to produce 5” square AAO templates with desired geometry • 6 months: 5” cellulose foils prepared by slot-die casting with thickness of 10–100 μm (σ<10%) and optical transmittance > 80%, as technology demonstration • 9 months: 5” cellulose foils with mechano-bactericidal surface nanostructures • 12 months: 8” templates with uniform patterns (σ<10% for nanowell diameter and density) • 15 months: 10” templates with uniform patterns (σ<10% for nanowell diameter and density) • 18 months: Manufacturing of large-area mechano-bactericidal cellulose foils • 21 months: Complete designing and building the roll-to-roll pilot manufacturing setup • 27 months: Rolls of 10” wide mechano-bactericidal foils as technology demonstration • 30 months: Bactericidal efficacy against > 7 strains of bacteria (ISO 22196) • 33 months: Performance verification in food models and hospital surfaces • 36 months: Complete set of documentations including comparison against competitors

R&D Plan

Aim 1. Fabrication of large-area micro-molding templates with precisely controlled geometry.

Rationale: A scalable, cost-effective manufacturing route for mechano-bactericidal polymer foils requires large-area templates with precisely defined nanopillar geometry. While our earlier silicon-based nanosphere-lithography process enabled exquisite control over pitch, diameter, and height, it is inherently expensive and limited to wafer-scale dimensions. To overcome these constraints, we will develop a scalable templating process based on electrochemical anodization of aluminum (AAO). By sputtering thin Al films onto low-cost substrates such as glass or Kapton and anodizing them in an industrial-scale plating bath, we can generate uniform nanoporous alumina templates over areas far larger than conventional silicon wafers. The anodization process naturally produces vertically aligned nanopores, and by orthogonally tuning the anodization voltage, Al film thickness, and post-anodization etching, we can independently control the nanowell pitch, diameter, and depth to match the optimum biomimetic geometry (~200 nm pitch, ~100 nm diameter, ~400 nm depth). This approach provides a practical pathway to large-area, low-cost templates suitable for continuous micro-molding process of biodegradable polymer foils to be developed in Aim 2.

Proposed research and development:

1. Identifying anodization conditions to produce AAO templates with desirable nanostructures: The first target is to generate AAO templates featuring nanowells with diameter ~100 nm, pitch ~200 nm, and depth ~400 nm, the optimum geometry justified in our previous studies. **(1)** Since we choose to deposit Al films on insulating and chemically inert glass or Kapton substrates and then fully oxidize the Al in anodization, the nanopore depth is self-limited by the Al film thickness. Due to the size expansion in oxidation, a 200–300 nm thick Al film will ensure final oxide thickness in the range of 300–500 nm. **(2)** The anodization conditions then need to be optimized to accurately control the nanowell pitch. The interpore pitch depends on the anodizing potential applied (**Fig. 6A**), as predicted by the stress-driven plastic flow transport model. We will optimize this hard anodization process (applied potential > 150 V to get pitch ~ 250 nm), with a focus on improving feature uniformity. Specifically, additives such as ethanol and ethylene glycol can improve the heat dissipation and increase the electrical resistivity of the electrolyte, which will increase the breakdown voltage and thus reduce the defect density in the AAO template while preserving the desired nanowell pitch. **(3)** Finally, the nanowell diameter needs to be controlled at around 100 nm, which will be accomplished mainly relying on the additional isotropic etching in phosphoric acid post anodization (**Fig. 6B**). With the nanowell depth, pitch, and diameter independently controlled by the Al film thickness, anodization voltage, and post-anodization etching time, we can fine tune the template topologies to further improve the performance of formed polymer films, in a closed loop integrating all three specific aims (i.e., template design–foil fabrication–performance evaluation).

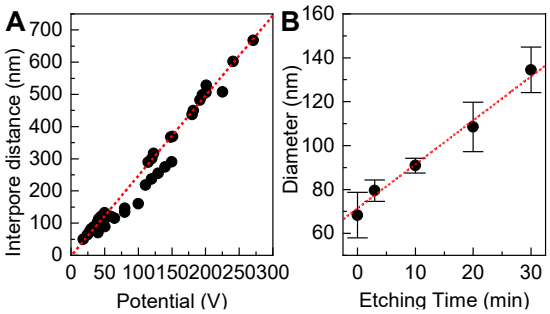

Fig. 6. Dependence of AAO structure on process parameters. Controlling (A) pitch via potential, and (B) diameter via etching.

2. Scaling up the fabrication of AAO templates with excellent uniformity over a large area: Our next target is to generate large-size AAO templates toward pilot manufacturing through collaboration with our SME partners. Aluminum anodization is a well-established industry-scale process of forming porous alumina surface coating on Al parts for enhanced corrosion resistance or decoration purposes (**Fig. 7A**). However, it is challenging to prepare AAO templates with excellent structural uniformity over a large area. We will tackle this uniformity challenge via optimizing the **(1) surface preparation** and **(2) electrochemical bath design**. We will systematically evaluate alkaline and acid cleaning, solvent degreasing, and pre-anodization chemical polishing to remove contaminants and ensure homogeneous electrochemical reactions across the entire substrate. More importantly, we will custom-build an electrochemical bath (**Fig. 7B**). We will innovatively apply the design of dual paddle plating cell used in precise semiconductor manufacturing.

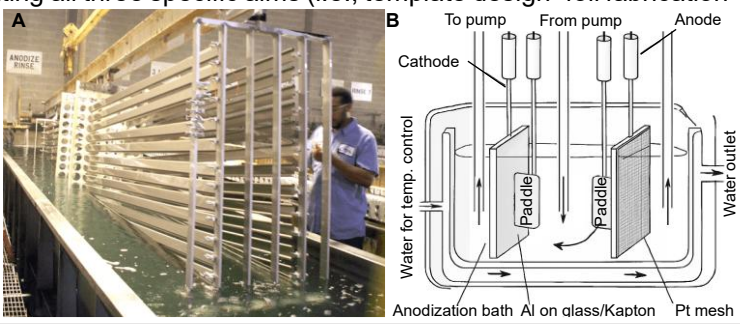

Fig. 7. Scaling up the fabrication of AAO templates. (A) Commercial bath**.** (B) Customized bath design.

Paddles are added close to the electrode surface to provide fluid agitation, ensuring **(1)** *uniform and efficient material transport* even for large-area substrates. The design and placement of anode and cathode fixtures, e.g., cathode shape and size, anode-to-cathode spacing, and electrical connections, will be optimized to ensure a **(2)** *uniform current distribution*. Meanwhile, we will incorporate dual bath circulation to improve **(3)** *concentration and temperature uniformities*. Uniformity will be quantitatively assessed using our machine-learning-assisted SEM image analysis pipeline, which can automatically extract nanowell dimensions from tens of thousands of features across a full substrate. This closed-loop feedback will guide iterative improvements in bath design and process conditions. The final deliverable for **Aim 1** is a pilot-scale anodization system capable of producing AAO templates with the required geometry and uniformity on 10-inch × 10-inch substrates, with 5-inch and 8-inch templates serving as intermediate milestones.

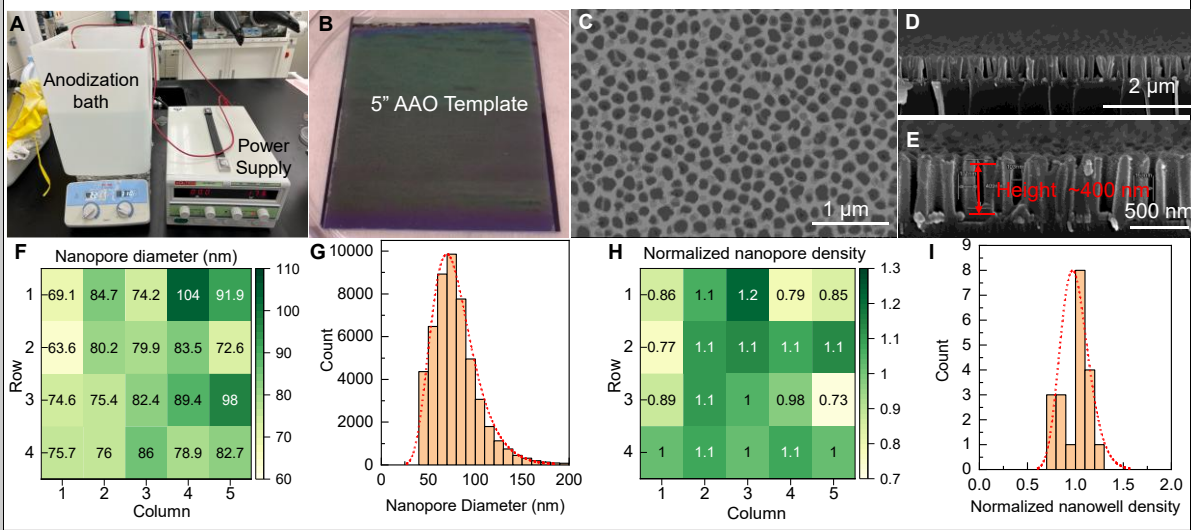

Fig. 8. Preliminary results on forming AAO templates. (A) Photo of the lab-scale anodizing setup. (B) 5" AAO template. (C-E) Top view (part C), cross-sectional SEM (part D) and magnified view (part E) of the AAO template. (F-I) Mapping of key geometric parameters, including nanowell diameter (part F) and the associated histogram (part G) as well as nanowell density (part H) and the histogram (part I).

Preliminary results: Our preliminary work demonstrates the feasibility of this approach. Using a lab-scale anodization setup (**Fig. 8A**), we produced 5-inch AAO templates under 200 V in 1 M H_3PO_4, achieving nanowells with ~100 nm diameter, ~180 nm pitch, and ~400 nm depth (**Fig. 8B–E**). The resulting pores exhibit smooth, vertical sidewalls—critical for defect-free demolding during polymer micro-molding. Using our ML-assisted analysis pipeline, we evaluated more than 50,000 nanowells across the substrate and confirmed excellent uniformity in both diameter and density (**Fig. 8F–I**). These results validate the technical foundation for scaling AAO template fabrication to even larger areas in collaboration with our SME partners in this project.

Aim 2. Scalable micro-molding processes to manufacture large-area, low-cost bio-degradable and sustainable foils featuring optimum mechano-bactericidal nanopillars on surface.

Rationale: To translate biomimetic mechano-bactericidal nanostructures into commercially viable antimicrobial films, we must develop a scalable, high-throughput manufacturing process that is compatible with low-cost, biodegradable polymers. Cellulose is an ideal candidate: it is renewable, widely available, biodegradable, biocompatible, and already used extensively in food and medical packaging. It also offers excellent mechanical strength, chemical stability, and barrier properties. The goal of this aim is to establish a continuous micro-molding process that can replicate the optimized nanopillar geometry onto large-area cellulose foils using the AAO templates developed in **Aim 1**. This will enable sustainable, low-cost antimicrobial films suitable for disposable packaging and hygiene-critical applications.

Proposed research and development:

1. Developing a micro-molding process to prepare cellulose foils with biomimetic mechano-bactericidal surface nanostructures: We will begin by formulating cellulose precursor solutions through dissolution of cellulose powders (derived from wood pulp or cotton) in the ionic liquid N-methyl-morpholine N-oxide (NMMO) (**Fig.9**). This precursor will be deposited onto large-area AAO templates using slot-die coating, a highly scalable, industry-standard technique that provides

excellent material utilization, precise thickness control, and compatibility with roll-to-roll manufacturing. The slot-die system uses a syringe pump to regulate precursor flow rate and an applicator head to distribute the solution uniformly across the coating width. We will optimize applicator geometry, including inlet design, manifold shape, land length, slot width, and shim configuration, to ensure stable, defect-free coating over AAO templates. After coating, the substrate will be immersed in a water bath to precipitate cellulose from the gel phase, forming a solid thin film. The film will then be delaminated/demolding from the template and dried.

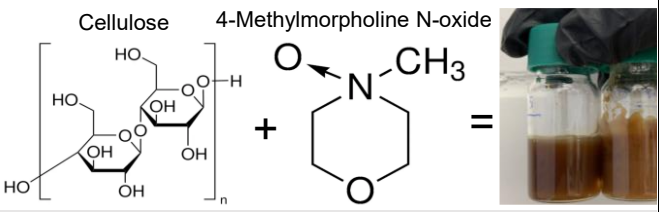

Fig. 9. Cellulose solution. Chemical formula of components and picture.

Key process parameters, including substrate speed and temperature, applicator-to-substrate gap, demolding pressure and speed, and cellulose concentration, will be systematically optimized. These parameters critically influence film thickness, mechanical properties (elastic modulus, elongation at break), optical transparency, barrier performance (oxygen and water permeability), and the fidelity of the replicated nanopillar structures. Iterative refinement will be guided by quantitative characterization of both film properties and nanostructure geometry.

2. Scaling up to roll-to-roll fabrication flow of antimicrobial cellulose foils: After optimizing the equipment design and process parameters on lab scale, we will rapidly move to scale up our process toward pilot manufacturing. We will build a continuous roll-to-roll flow (**Fig. 10A**). It will employ a pair of 300 mm wide rollers on a 500 mm wide platform for continuous operation (**Fig. 10B**). After dispersion of cellulose solution on a template (Kapton foil covered by the AAO template prepared in **Aim 1**, which is further laminated on top of a stainless steel belt; the Kapton film can be replaced to clean or regenerate the AAO template) by slot-die coating (**Fig. 10C**), the cellulose gel film and template will then go through an integrated inline oven (**Fig. 10D**) kept at ~120 °C, which decreases the film viscosity and thus promote the rapid diffusion of cellulose into the template nanowells. Afterward, the film stack will be introduced into a water bath for coagulation and washing (**Fig. 10E**). After drying by an air knife, a nip roller drives delamination of foils from template (**Fig. 10F**). Throughout the project, our design will be continuously refined to increase manufacturing throughput and yield, leveraging the expertise of the SME collaborator.

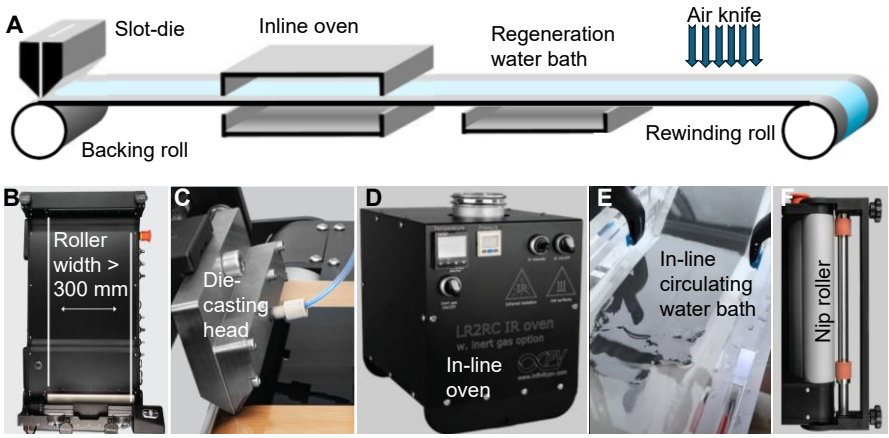

Fig. 10. Roll-to-roll manufacturing. (A) Schematic. (B-E) Photos of platform (B), slot-die head (C), inline oven (D), bath (E), and nip roller (F).

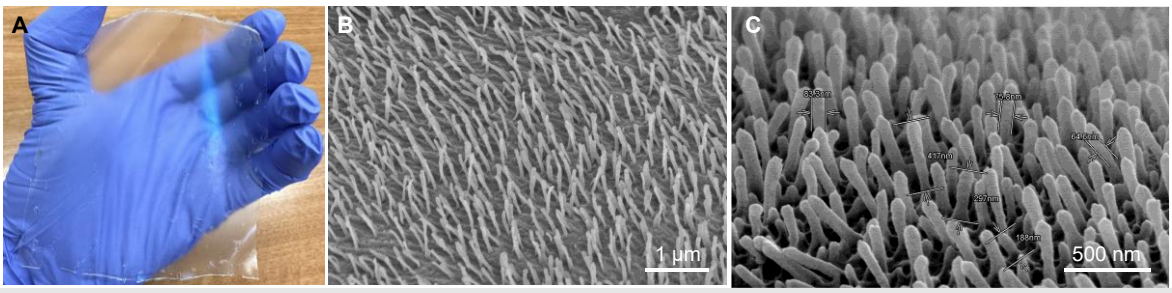

Fig. 11. Cellulose foils prepared with AAO templates. (A) Optical image of a 5” foil. (B) SEM showing the nanopillar-array structures on this cellulose foil generated from a AAO template. (C) SEM showing the nanopillar generated from an AAO template that has been repetitively used for five times.

Preliminary results: Our preliminary studies confirm the feasibility of this approach. Using the 5-inch AAO templates developed in Aim 1 (**Fig. 8B**), we successfully fabricated cellulose foils with the desired mechano-bactericidal nanopillar geometry—~200 nm pitch, ~100 nm diameter, and ~400 nm height (**Fig. 11A–B**). Importantly, the AAO templates can be reused at least five times (**Fig. 11C**) without degradation of nanostructure fidelity, demonstrating their robustness for continuous manufacturing. These results validate the technical foundation for scaling to roll-to-roll production in collaboration with our SME partners.

Aim 3. Characterization and benchmark of antimicrobial performance of mechano-bactericidal cellulose foils following the ISO protocols and in food models and hospital environment.

Rationale: To support regulatory approval, customer adoption, and market entry, it is essential to rigorously validate the antimicrobial performance, durability, and real-world effectiveness of our mechano-bactericidal cellulose foils. While our laboratory studies have demonstrated strong bactericidal activity, commercial deployment requires standardized benchmarking under internationally recognized protocols and performance verification in realistic use environments. This aim establishes the scientific and regulatory foundation for translation by combining ISO-compliant microbiological testing with evaluations in food models and hospital settings.

Proposed research and development.

1. *In vitro* microbiological tests following established protocols for benchmarking: We will quantitatively assess the antimicrobial efficacy of our cellulose-based mechano-bactericidal films using the ISO 22196 standard, the global benchmark for evaluating antibacterial activity on non-porous surfaces. All experiments will include internal controls, cross-comparisons with commercial antimicrobial films, and statistical rigor (triplicate samples from three independent experiments; unpaired two-tailed Student's t-test or ANOVA; significance threshold $p \leq 0.05$).

Following ISO 22196 protocol (**Fig. 12**), common pathogenic Gram-negative and Gram-positive strains, including *Escherichia coli, Pseudomonas aeruginosa, Klebsiella pneumoniae, Salmonella enterica, Staphylococcus aureus, Listeria monocytogenes, Clostridium botulinum*, etc., will first be incubated in cell-culture medium to achieve exponential-phase growth. The cell suspensions are then diluted to reach a cell density above 10^6 CFUs per mL. Each film undertest (5×5 cm^2, including our mechano-bactericidal films and planar foils as controls) will be inoculated with 0.4 mL cell dispersion and covered with a sterile planar film to ensure maximum contact and prevent drying. Films will be incubated at 37 °C with >90% relative humidity for 24 hours. Bacteria will then be thoroughly rinsed from the test surfaces. Collected bacterial washes will be diluted, plated onto LB agar plates, and incubated to determine the number of CFUs, quantitatively expressed as Log_{10} reduction of the bacterial burden compared to planar control foils. To assess durability and lifetime, we will also evaluate films subjected to simulated use conditions, including repeated finger pressing, washing-drying cycles, and mechanical handling. These tests will determine whether the nanopillar structures retain their bactericidal efficacy under realistic wear, an essential requirement for commercial packaging and hygiene applications.

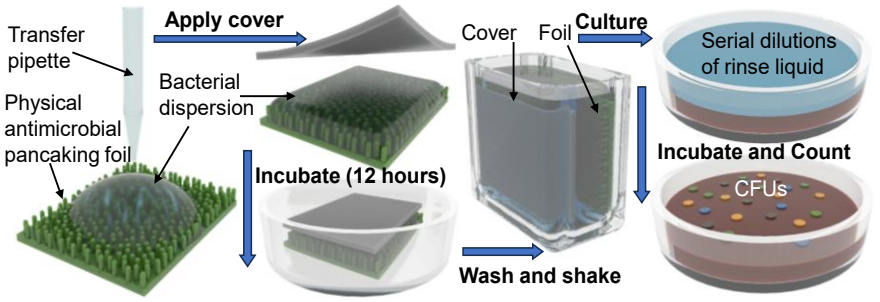

Fig. 12. Schematic of the ISO 22196 protocol.

2. food models and field test: To demonstrate real-world relevance and generate data that resonate with end-users, we will evaluate antimicrobial performance in food models and hospital environments.

Food models: We will test chicken breast, salmon fillets, and cheese, which are representative substrates for food packaging. Each sample will be wrapped with either our mechano-bactericidal foil or planar controls and sealed in vacuum bags to ensure intimate contact. Samples will be stored at 5 °C and room temperature for 3, 6, and 9 days. At each time point, we assess spoilage qualitatively (odor, appearance) and quantitatively by homogenizing samples in peptone water and performing standard plate counts. These results will demonstrate the ability of our films to suppress bacterial growth and extend shelf life, providing compelling evidence for food manufacturers and retailers.

Hospital field test: After obtaining ethical approval, we will apply our mechano-bactericidal foils to toilet seats in a high-traffic medical ward, with untreated plastic surfaces as controls. Weekly sampling over 10 weeks will be performed using sterile saline-moistened swabs. Swabs will be vortexed in

	sterile saline, and the recovered bacteria will be quantified. Typical aerobic bacterial loads on untreated surfaces range from 30–200 CFU/cm²—sufficient to detect meaningful reductions. This study will demonstrate the technology's ability to reduce microbial contamination in hygiene-critical environments, supporting adoption by hospitals and public facilities. **Preliminary results:** The cellulose foils with surface nanopillar arrays defined by the AAO templates are very effective against *E.coli* as tested following the ISO 22196 protocol, showing >10^5 times reduction of bacterial viability compared to the planar controls, while commercial foils and those with micron-scale hydrophobic surface patterns exhibited no bactericidal activities.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The university will lead the scientific and technical development of the biomimetic mechano-bactericidal technology, building on its foundational discoveries in nanopillar design, polymer–nanostructure template interactions, and antimicrobial mechanisms. Its primary responsibilities include optimizing the geometry of the nanopillar arrays, developing and refining the AAO template fabrication process, and establishing the micro-molding and slot-die coating conditions needed to replicate high-fidelity nanostructures on biodegradable cellulose foils. The university will also conduct comprehensive evaluations of antimicrobial efficacy and safety, including ISO 22196 benchmarking, durability testing, and performance validation in food models and hospital environments. In addition, the university will provide access to advanced nanofabrication, microscopy, and characterization facilities, and will generate the scientific datasets, mechanistic insights, and performance metrics required to support regulatory submissions and product positioning. Through iterative experimentation and data-driven optimization, the university will ensure that the technology achieves the required levels of efficacy, reproducibility, and safety for commercial translation.
Expected R&R of Korean SME Partner	The SME partners will play a central role in scaling, engineering, and commercializing the mechano-bactericidal film. Their contributions include co-developing the large-area AAO templating system, designing and refining the roll-to-roll manufacturing line, and ensuring that the process meets industrial requirements for throughput, cost, and yield. SME engineers will assist in optimizing slot-die coating hardware, fluid-delivery systems, oven and bath configurations, and delamination mechanics to enable continuous production of antimicrobial foils. They will also lead manufacturability assessments, reliability testing under industrial conditions, and early pilot-scale production runs. Beyond engineering, the SME partners will contribute market intelligence, regulatory strategy, and customer-driven performance requirements, helping to align the technology with the needs of food manufacturers, packaging suppliers, hospitals, and consumer-hygiene markets. Their resource commitments may include engineering personnel, prototyping equipment, pilot-line access, and support for ISO-compliant testing. The SME partners will ensure that the technology is not only scientifically validated but also economically viable and scalable.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint R&D project is expected to deliver a fully validated, scalable manufacturing platform for producing large-area, biodegradable mechano-bactericidal cellulose foils, establishing a new class of chemical-free antimicrobial materials for food packaging, healthcare, and hygiene-critical applications. Technologically, the project will achieve several first-in-class milestones: large-area AAO templates with precisely controlled nanoscale geometry; high-throughput roll-to-roll micro-molding of nanopillar-structured cellulose films; and ISO-certified antimicrobial performance that surpasses existing commercial chemical-release films. These achievements will position the partners to introduce a disruptive, environmentally safe alternative to biocide-based antimicrobial coatings, addressing global concerns around packaging toxicity, antimicrobial resistance, and plastic waste. Commercially, the project will enable the SME partners to establish proprietary manufacturing capabilities, accelerate product launch, and capture emerging market opportunities in sustainable packaging and medical-grade antimicrobial surfaces.